Additional README Corrections

1. Analytical Task 1 – Model Specification (Required)

The current Model Specification section incorrectly describes the final-stage model as a **causal forest**.

Current wording:

Final stage | ForestDRLearner (causal forest)

This is technically incorrect. My implementation uses **ForestDRLearner**, whose final stage is a **Random Forest regression model**, not a Causal Forest (CausalForestDML).

The table should instead describe the final stage as something similar to:

Component	**Model**	**Purpose**
Outcome model	RandomForestRegressor	Estimates baseline ED risk
Treatment model	LogisticRegressionCV	Estimates treatment propensity
Final-stage regression	Random Forest regression model (within ForestDRLearner)	Learns the relationship between doubly robust pseudo-outcomes and individualized treatment effects

The phrase **"causal forest"** should be removed from this section entirely.

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2. Analytical Task 1 – Model Specification Description (Required)

The paragraph describing how ForestDRLearner works should also be updated.

Current wording suggests:

ForestDRLearner constructs pseudo-outcomes and fits a final-stage causal forest.

Instead it should describe the process as:

ForestDRLearner first constructs doubly robust pseudo-outcomes using the outcome models and propensity model. A final-stage Random Forest regression model is then trained on these pseudo-outcomes to estimate individualized treatment effects for new members.

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3. Analytical Task 1 – Mermaid Workflow Diagram (Required)

The current Mermaid diagram is too similar to the Causal Forest workflow and does not accurately describe how a Doubly Robust Learner operates.

The workflow should clearly illustrate:

Training Data

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Outcome Models

│

▼

Predicted μ_1 and μ_0

│

▼

Initial Treatment Effect ($\mu_1 - \mu_0$)

│

▼

Residual Corrections

• Treated correction

• Untreated correction

│

▼

Construct Doubly Robust Pseudo-Outcome

│

▼

Random Forest Regression

│

▼

Estimated Treatment Effect

│

▼

Benefit Score

│

▼

Rank Members

This better explains the DR learner rather than attempting to mirror the Causal Forest workflow.

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4. Analytical Task 1 – "What a Doubly Robust Learner Estimates" (Recommended)

I would expand this explanation slightly.

Rather than simply stating that pseudo-outcomes are constructed and a forest is fit, explicitly explain the three components of the pseudo-outcome:

- Initial treatment-effect estimate from the difference between predicted treated and predicted control outcomes.
- Treated correction term using the treated residual weighted by inverse propensity.
- Control correction term using the untreated residual weighted by inverse propensity.

Then explain that the sum of these three terms forms the doubly robust pseudo-outcome, and that a final Random Forest regression model is trained on these pseudo-outcomes to estimate individualized treatment effects.

This is one of the most important conceptual sections of the README and should clearly distinguish the DR learner from the X-learner and Causal Forest.

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5. Analytical Task 3 – Propensity and Overlap Checks (Recommended)

The propensity overlap diagnostics are already fully presented in the Causal Forest notebook and README.

Since the same shared propensity scores are reused here, there is no need to duplicate the overlap histogram and detailed discussion.

Instead, briefly state that:

- Shared propensity scores generated by the X-learner workflow are reused.
- The overlap diagnostics are identical to those shown in the Causal Forest analysis.

This keeps the README shorter and avoids unnecessary repetition.

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6. Analytical Task 3 – Pseudo-Outcome Diagnostics (Required)

The current notebook is not actually plotting the doubly robust pseudo-outcomes.

Instead, it is plotting the estimated treatment effects returned by:

dr_model.effect(X_train)

These are estimated CATEs, not the raw doubly robust pseudo-outcomes.

Therefore, one of the following changes should be made:

Option A (preferred)

Actually reconstruct and display the true doubly robust pseudo-outcomes.

Option B

Rename the section to something like:

Training-Set Treatment Effect Distribution

rather than:

Pseudo-Outcome Diagnostics.

The current title is technically misleading.

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7. Analytical Task 4 – Treatment Effect Distribution (Required)

The treatment-effect distribution should be moved from Task 3 into Task 4.

Task 4 is specifically focused on **Treatment Effect Analysis**, making it the appropriate location for the effect distribution figure and discussion.

The training-set effect distribution currently shown in Task 3 is unnecessary.

Only the test-set treatment-effect distribution should be presented.

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8. Analytical Task 5 – Risk Tier Versus Benefit Group (Recommended)

The formatting of this section should match the Causal Forest README exactly.

The layout, table formatting, and chart presentation should be consistent across both READMEs.

If necessary, refer to the generate_readme_tables.py implementation used by the Causal Forest notebook so that the presentation style is identical.

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9. Clarify Shared Models (Recommended)

Add a brief clarification explaining that:

- The **propensity scores are shared** across the X-learner, Causal Forest, and Doubly Robust workflows.
- The **outcome models are not shared**. ForestDRLearner estimates its own nuisance outcome models internally during cross-fitting before constructing the doubly robust pseudo-outcomes.

This helps explain why the methods remain comparable while still estimating treatment effects independently.

Priority Summary

Required

- Remove all references to a "final causal forest."
- Update the Model Specification table.
- Rewrite the Mermaid workflow to accurately reflect the DR algorithm.
- Move the treatment-effect distribution from Task 3 to Task 4.

- Correct the pseudo-outcome section (either compute the true pseudo-outcomes or relabel the section).

Recommended

- Remove the redundant propensity overlap diagnostics.
- Match the Task 5 formatting to the Causal Forest README.
- Clarify that propensity scores are shared, but outcome models are estimated independently by ForestDRLearner.